

car or objects around the car, fastest and safe route to destination, and capability to control the driving system. It is also important to know when and how to perform required actions.

Components required for autonomous car design

Autonomous cars being tested today make use of a combination of laser illuminating detection and ranging (lidar), radio detection and ranging (radar), sound navigation and ranging (sonar), cameras, GPS and inertial measurement units (IMUs) along with 3D road maps. These sensors and systems work together, and data is analysed in real time to manipulate the steering, acceleration and braking.

Radar sensors help monitor the position of nearby vehicles. Ultrasonic sensors assist vehicles while parking. Use of both types of sensors has resulted in lidar technology. Lidar sensors help detect the edges of roads and identify lane markings by bouncing pulses of light off the car's surroundings. These also alert cars to nearby objects, including other cars, pedestrians, bicyclists and other obstacles.

Lidar technology is also used to build a 3D map to enable the car to see its surroundings and spot potential hazards by determining the size and distance of all things around it. It is great at capturing information in various types of ambient light—whether night or day, sunny or cloudy.

Lidar and cameras, together with radar and GPS antennae, work to make the car perceive its surroundings and allow it to determine its position. Cameras detect traffic lights, read road signs and road markings, and keep track of other vehicles, while also looking out for pedestrians, bicyclists, cars and other obstacles. These may, however, face difficulty in shadows or poor lighting conditions. A combination of ultrasonic, radar, cameras, GPS antennae and lidar sensors digitally map out the road ahead of an autonomous car and help it see where it is going.

While Google continues to rely on lidar, Tesla strongly believes that camera vision will ultimately prove superior. Ziv Aviram, chief executive officer, Mobileye, agrees, "Camera vision is simply more sophisticated

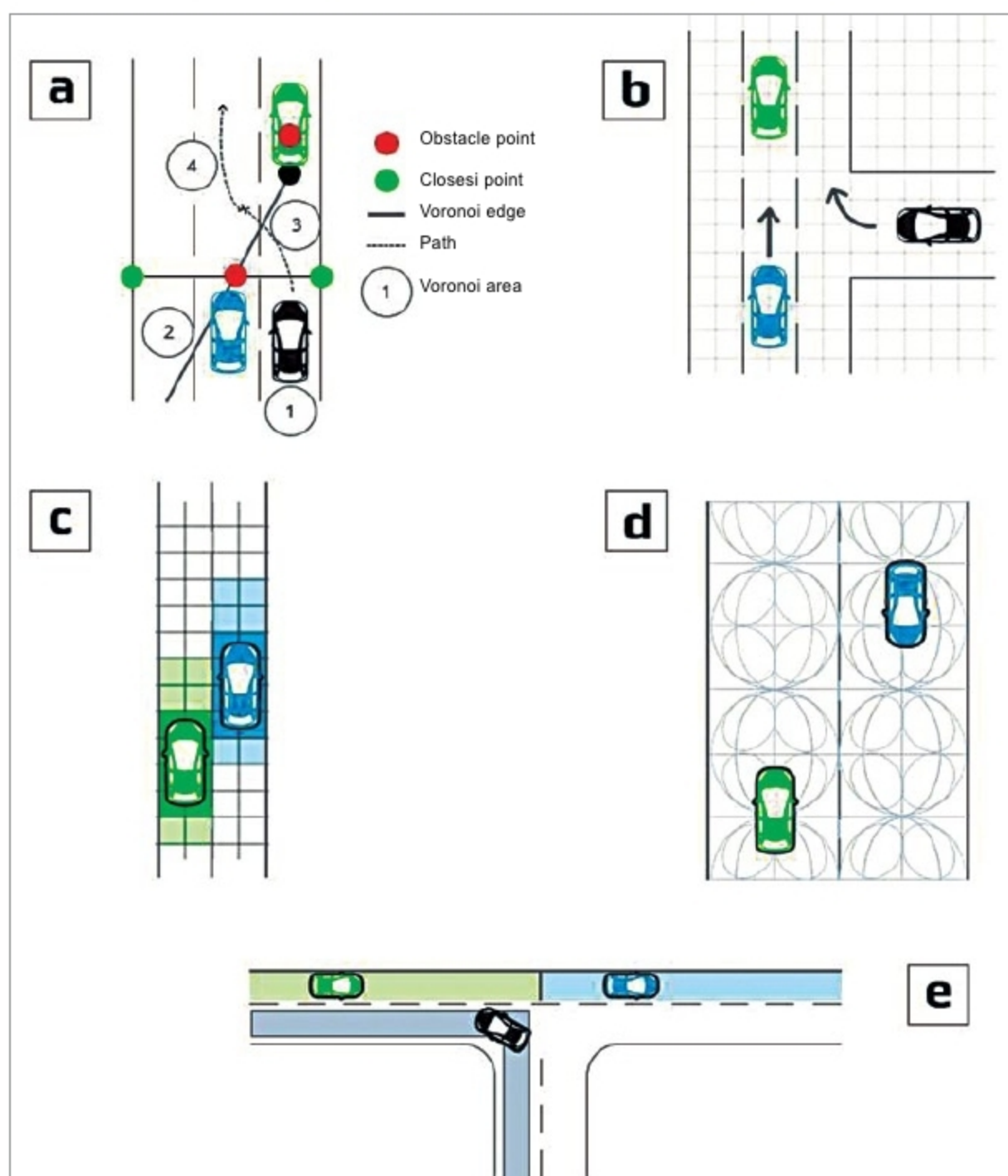


Fig. 1: Path planning technology (Credit: <https://dzone.com>)

than lidar, and offers much higher resolution data."

Satyarth Praveen, research engineer, Hi-Tech Robotic Systemz, adds, "It depends on the use case. Lidar has a resolution factor as well. Since it makes measurement based on reflected light, lidar makes the device scene-dependent as well. Meanwhile, cameras are more dependent on the software being used to understand the captured scene. Since lidar is immune to atmospheric variations, missed detection is less of an issue with it. So just the selection of a device cannot guide towards what works better; there are a lot of other things that go along with it."

Path planning technology

Hard-coded rules, obstacle avoidance algorithms, predictive modelling and smart object discrimination between different vehicles on the road help the

software follow traffic rules and navigate through obstacles. To stay on the road, avoid collisions and obey traffic laws, autonomous cars rely on path planning approaches including predictive control and behaviour-based models. Algorithms process the information gathered by the autonomous car technology to enable the cars to react accordingly to variables such as traffic, road construction, detours and varying weather conditions.

Some algorithms used for finding corridors and spaces are:

- Voronoi diagram algorithm, which generates paths that maximise the distance between a vehicle and surrounding obstacles
- Occupancy grid algorithm, which calculates risk and feasibility based on the presence of obstacles, and lane and road boundaries
- Driving corridors algorithm,